Chapter 5 — Python cleaning tools

Methods become operational through libraries

Learning objectives

- Perform dropna, fillna
- Drop_duplicates in pandas, apply StandardScaler and OneHotEncoder from scikit-learn
- Outline an end-to-end cleaning walkthrough

pandas for cleaning

- Pandas is the default toolkit for tabular cleaning
- Example 5.57 shows dropna and drop_duplicates in one script
- Example 5.59 walks an end-to-end cleaning template from raw extract to model-ready frame

Example 5.57 — Pandas dropna and drop duplicates

- Example 5.57 — hands-on module
- The minimal cleaning API surface
- Explore the chapter example module
- View files: `modules/chapter5/example57/`

Example 5.57 — listing

```
import pandas as pd
df = pd.read_csv("data.csv")
df = df.dropna()  # Removing rows with missing values
df = df.drop_duplicates()  # Removing duplicate rows
```

scikit-learn for preprocessing

- Scikit-learn provides fit-transform estimators
- Example 5.60 standardizes age and salary; Example 5.61 one-hot encodes gender
- Fit on training data only to prevent leakage

Example 5.59 — Pandas end-to-end cleaning walkthrough

- Example 5.59 — hands-on module
- Example 5.59 ties issues to fixes in one narrative
- Explore the chapter example module
- View files: `modules/chapter5/example59/`

Example 5.59 — listing

```
import pandas as pd

# Load data
df = pd.read_csv("customers.csv")

# 1. Handling Missing Data
df['age'] = df['age'].fillna(df['age'].mean())
df = df.dropna(subset=['name', 'email'])

# 2. Removing Duplicates
df = df.drop_duplicates(subset=['email'])

# 3. Standardizing Column Formats
df['name'] = df['name'].str.title()

# 4. Encoding Categorical Data
# ...
```

When to use which layer

- Use pandas for exploratory fixes and bespoke string parsing
- Use sklearn transformers when the same preprocessing must repeat across folds and
- Keep business rules and unit conversions visible in notebooks or pipeline comments

Takeaways

- Pandas repairs tables; sklearn standardizes repeatable transforms
- Fit transformers on training splits
- Example modules provide runnable templates for this chapter

Next

- Complete the quiz for this part
- Continue to pipelines, automation, visualization, and a brief R contrast